Lab 3: Information Retrieval

Session 2. Search for documents with Whoosh.

Goal

The lab session aims to provide students with comprehensive, hands-on experience in information retrieval techniques using Python's Whoosh library. Students will develop skills in constructing and managing document indices, implementing search strategies across multiple fields, and applying advanced filtering. Through practical exercises, students will learn to define complex queries and perform searches on the indexed document collection. The lab focuses on transforming raw textual data into efficiently searchable and retrievable information, emphasizing real-world applications.

Bibliography

- https://pypi.org/project/Whoosh/
- https://whoosh.readthedocs.io/en/latest/index.html

INTRODUCTION

Whoosh is a library containing classes and functions for indexing text and searching on the created index. It allows the development of custom search engines.
Text indexing, the information stored for each term in each field, the parsing of search queries, the types of the allowed queries, the ranking algorithms, etc., are customizable, replaceable, and extensible.

- Whoosh uses only Python, so only the Python interpreter is needed.
- Whoosh uses, by default, Okapi BM25F (https://es.wikipedia.org/wiki/Okapi_BM25) as a ranking function, but it can be customized.
- Whoosh creates small indexes compared to many other libraries.
- All text indexed in Whoosh must be in Unicode.
- Whoosh allows you to associate and store Python objects with indexed documents.

STEPS TO RETRIEVE DOCUMENTS.

1. Open the index from the directory. You can also specify the name of the index (if you have changed the default name):

```
ix = open_dir("index_dir")
```

2. A searcher object has to be created. Once the searches are finished, this object must be closed:

```
searcher = ix.searcher( )
          ...
searcher.close( )
```

3. The searcher object needs an object of type Query. Also is required to define a parser for the query content:

```
parser = QueryParser("content", ix.schema)
query = parser.parse("the_query_text")
```

4. We can do a manual parse or create our Query:

```
from whoosh.query import *
          myquery = And([Term("content", "term1"),
          Term("content", "term2")])
```

5. The search is launched, and the results are returned:

```
          results = searcher.search(query)
          results[0]
```

EXAMPLE 1

```
from whoosh.index import open_dir
from whoosh.qparser import QueryParser

ix = open_dir("path_to_the_index_directory")
# Using the "with" python statement it is not required to close
the searcher object
with ix.searcher() as searcher:
        parser = QueryParser("content", ix.schema)
        query = parser.parse("text_of_the_query")
        results = searcher.search(query)
        print(results[0])
```

The main technical details on the most important functionality of Woosh for performing the search on the indexed collection of documents can be found in the document "LNR2025-Woosh-S.pdf".

ACTIVITY 1

Write a Python program to search for information using the indexes created in the previous session. The program must receive as first argument the directory where the index is located and, optionally, two more arguments to determine the execution setting:

```
python LNR_searcher.py index_directory [-q=query] [--extend]
```

-q=query: It allows you to search from the command line. The program will show the result of the query and terminates. If this argument is not used, the program enters a keyboard query request loop and shows the results until an empty query is entered.

--extend: It shows only the first result of the search, but in an extended form, it must show all the stored fields. If this argument is not used, one news per line will be displayed with the date and title. The total number of news retrieved will be shown in both cases.

EXAMPLE 2

```
python LNR_searcher.py ./index -q Emigration

Query: content:emigration
-> (2019-02-23 00:00:00) Putin has Provoked an Emigration Equal in Size and Importance to the One the Bolsheviks Did
-> (2019-04-02 00:00:00) EU elections: Eastern, southern Europeans dread emigration more than immigration
-> (2019-11-11 00:00:00) Gaza Specialist Doctors Emigrate Due to Israel Siege
...
...
...
-> (2019-11-10 00:00:00) The Russians Who Went West: A Lost Generation of Emigres
```

-> (2019-03-13 00:00:00) How Jewish Elites Use Non-White Immigration to Undermine Western Nations (VIDEO)
-> (2019-08-31 00:00:00) Can Private Cities Help Solve the Refugee Crisis?
-> (2019-10-31 00:00:00) Lebanon's "Revolution" Is Without Revolutionary Ideology
==========
29 results

EXAMPLE 3

python3 LNR_searcher.py ./index -q Emigration --extend

Query: content:emigration
Date: 2019-02-23 00:00:00
Title: Putin has Provoked an Emigration Equal in Size and Importance to the One the Bolsheviks Did
Source: windowoneurasiablog
Collection: 2019-09-03 23:52:49
Published: 2019-02-23 06:41:00
Category_level_1: society
Category_level_2: emigration
Url: http://windowoneurasia2.blogspot.com/2019/02/putin-has-provoked-emigration-equal-in.html
…
==========
29 results

EXAMPLE 4

python LNR_searcher.py ./index

Enter a query: title:Emigration OR title:muslim

Query: (title:emigration OR title:muslim)
-> (2019-07-03 00:00:00) Eastern Europe's emigration crisis
-> (2019-11-12 00:00:00) Venezuelans Who Return From Emigration and Stories Nobody Tells
-> (2019-03-29 00:00:00) Government will not restrict emigration to address brain drain issue
...
...

```
...
-> (2019-01-18 00:00:00) Kenya: Devout Muslim terrorists who attacked hotel and murdered 14 say they were targeting “Zionist Jews”
-> (2019-02-06 00:00:00) Politician who said &apos;the Quran is poison&apos; announces he has become a Muslim after trying to write anti-Islam book
-> (2019-06-28 00:00:00) Dalai Lama sparks ‘sexism’ outrage with joke about appointing good looking female successor day after saying Europe ‘will become Muslim if refugees don’t leave’
=================
89 results
```

Enter a query:

ACTIVITY 2

Enhance the functionality to conduct the specified searches by printing out **the news content** that satisfies the queries when the "*extend"* option is selected. The search mechanism should retrieve all news matching, not just the exact queries, but also disregard distinctions between plurals and singulars, third-person verbs, and case sensitivity. For instance, when the query *'love'* is given the program should fetch all news containing variations like *'love', 'LoVE', 'loves', 'lover', 'Lover', 'lovers*', etc. Adjust the schema, the indexer, and the searcher (if needed) accordingly to facilitate the implementation of these new searches.

EXAMPLE 5

python LNR_searcher.py ./index -q “title:Emigration OR title:muslim”

```
Query: (title:emigr OR title:muslim)
-> (2019-07-03 00:00:00) Eastern Europe's emigration crisis
-> (2019-05-31 00:00:00) Pope Highlights 'Sacrifices' Of Romanian Emigrants
-> (2019-04-04 00:00:00) Record Number of Russians Want to Emigrate – Gallup
-> (2019-11-11 00:00:00) Gaza Specialist Doctors Emigrate Due to Israel Siege
```

...
...
...
-> (2019-03-17 00:00:00) All the religious massacres you didn't hear about over the past year have one thing in common... Christians were killed instead of Muslims
-> (2019-02-06 00:00:00) Politician who said 'the Quran is poison' announces he has become a Muslim after trying to write anti-Islam book
-> (2019-06-28 00:00:00) Dalai Lama sparks ‘sexism’ outrage with joke about appointing good looking female successor day after saying Europe ‘will become Muslim if refugees don’t leave’
==========
160 results